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Herpes Zoster

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Abstract

A 65-year-old man presents with a rash of 2 days duration over the right forehead with vesicles and pustules, a few lesions on the right side and tip of the nose, and slight blurring in the right eye. The rash was preceded by tingling in the area and is now associated with aching pain. How should this patient be evaluated and treated?

THE CLINICAL PROBLEM

Primary infection with varicella-zoster virus (VZV) results in chickenpox, manifested by viremia with a diffuse rash and seeding of multiple sensory ganglia where the virus establishes life-long latency. Herpes zoster is caused by reactivation of latent VZV from cranial nerve or dorsal root ganglia with spread of virus along the sensory nerve to the dermatome. There are more than 1 million cases of herpes zoster in the United States each year with an annual rate of 3 to 4 cases per 1000 persons. Studies suggest that the incidence of herpes zoster is increasing.¹ Unvaccinated persons who live to age 85 have a 50% chance of developing herpes zoster. Up to 3% of patients with zoster require hospitalization.

The major risk factor for herpes zoster is increasing age. With increasing time after varicella, there is a reduction in the level of T cell immunity to VZV² which, unlike virus-specific antibodies, correlates with protection from zoster. The risk is higher in women than men, in whites than blacks, and in persons with a family history of herpes zoster.³

Chickenpox that occurs in utero or early in infancy, at a time when the cellular immune system is not fully matured, is associated with herpes zoster in childhood.

Immunocompromised persons with impaired T cell immunity, including organ and hematopoietic stem cell transplant recipients, persons receiving immunosuppressive therapy, and patients with lymphoma, leukemia, or HIV infection, are at increased risk of developing herpes zoster and having more severe disease.

Postherpetic neuralgia (PHN), or pain persisting after the rash has resolved (often defined specifically as pain persisting 90 days or more after onset of rash), is a feared complication of herpes zoster. The pain may persist for many months or even years; it may be severe and

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interfere with sleep and activities of daily living, resulting in anorexia, weight loss, fatigue, depression, withdrawal from social activities and employment, and loss of independent living. Depending on age and the definition used, ten to 50% of persons with herpes zoster develop PHN. The risk increases with increasing age (particularly over age 50) and is greater in persons with severe pain at the onset of herpes zoster or severe rash with a large number of lesions.

Various neurologic complications have been reported to occur with herpes zoster including Bell's palsy, the Ramsay-Hunt syndrome, transverse myelitis, transient ischemic attacks, or stroke.⁴ In addition, ophthalmologic complications of zoster occurring in the V1 distribution of the trigeminal nerve can include keratitis, scleritis, uveitis, or acute retinal necrosis (Table 1). Immunocompromised persons can develop additional complications including disseminated skin disease, verrucous skin lesions, acute or progressive outer retinal necrosis, chronic zoster, or infection with acyclovir-resistant VZV. In these patients the disease can involve multiple organs (e.g. lung, gastrointestinal tract, brain) and patients may present with hepatitis or pancreatitis several days before the rash appears.⁵

STRATEGIES AND EVIDENCE

EVALUATION

The rash of herpes zoster is dermatomal and does not cross the midline, consistent with reactivation from dorsal root or cranial nerve ganglia. The thoracic, trigeminal (Fig. 1A), lumbar, and cervical dermatomes are the most frequent sites of rash, although any area of the skin can be involved. In non-immunocompromised persons, a few scattered lesions outside the affected dermatome are not unexpected. The rash is often preceded by tingling, itching, and/or pain for 2 to 3 days, which can be continuous or episodic. Depending upon the location and severity, this prodromal pain may lead to misdiagnosis and costly testing. The rash begins as macules and papules which evolve into vesicles and then pustules (Fig. 1B). New lesions appear over 3 to 5 days, often with filling in of the dermatome despite antiviral treatment. The rash usually dries with crusting in 7 to 10 days. Some persons have pain in the absence of a rash, termed zoster sine herpete, which is difficult to diagnose and may lead to numerous unnecessary tests or procedures. Immunocompromised patients may have disseminated rashes with viremia and new lesions occurring for up to 2 weeks. The characteristics of pain associated with herpes zoster may vary. Patient may have paresthesias (e.g. burning or tingling), dysethesias (altered or painful sensitivity to touch), allodynia (pain associated with non-painful stimuli) or hyperesthesia (exaggerated or prolonged response to pain). Pruritis is also commonly associated with herpes zoster.

DIAGNOSIS

Most cases of herpes zoster can be diagnosed clinically, although direct immunofluorescence for VZV antigen or PCR for VZV DNA in cells from the base of lesions after they are unroofed may be needed for atypical rashes. In a study comparing PCR with other diagnostic methods, the sensitivity and specificity of PCR was 95% and 100% respectively, and, using immunofluorescent VZV antigen testing, was 82% and 76%, respectively.⁶ The most common condition mistaken for zoster is herpes simplex virus,

which can recur in a dermatomal distribution; accordingly, when patients present with “recurrent zoster,” atypical lesions, or are immunocompromised with disseminated skin lesions, specific testing for both VZV and herpes simplex virus is often useful. VZV has been demonstrated in saliva of persons with zoster, although such testing does not currently have a demonstrated role in clinical practice.⁷

PCR of the cerebrospinal fluid (CSF) has been used for diagnosis of CNS vasculopathy; demonstration of an increase in the ratio of anti-VZV antibody in CSF to that in the blood is more sensitive.⁴ PCR of the blood may be helpful for diagnosis of visceral zoster in immunocompromised persons who present with hepatitis or pancreatitis without a rash.⁵ PCR of blood or CSF for VZV has been used for diagnosis of zoster sine herpette.

TREATMENT AND PREVENTION

Antiviral Therapy—Antiviral therapy is recommended for herpes zoster for certain non-immunocompromised patients and all immunocompromised patients (Table 2). Other persons might also benefit from antiviral therapy, although their risk of complications from zoster is lower. Three guanosine analogs—acyclovir, valacyclovir, and famciclovir—have been licensed by the Food and Drug Administration (FDA) for treatment of herpes zoster (Table 3). Oral bioavailability and levels of antiviral drug activity in the blood are higher and more reliable in those receiving thrice daily valacyclovir or famciclovir than for acyclovir 5 times daily. This is important, because VZV is less sensitive than herpes simplex virus to acyclovir, valacyclovir, or famciclovir. These antivirals hasten resolution of lesions, reduce new lesion formation, reduce virus shedding, and decrease severity of acute pain (Table 3). For example, in the largest randomized, double-blind trial of acyclovir for zoster, oral acyclovir given within 47 hours of onset shortened the mean time for the last day of new lesion formation, loss of vesicles, and full crusting by 0.5 days, 1.8 days, and 2.2 days, respectively, compared with placebo.¹⁰ In another large trial, acyclovir reduced duration of virus shedding by 0.8 days compared with placebo.¹¹ In a meta-analysis of several randomized, controlled trials, antivirals did not significantly reduce the incidence of PHN¹⁹ and they are not approved for prevention of PHN by the FDA. Valacyclovir or famciclovir have been shown to be superior to acyclovir for reducing pain associated with herpes zoster in some studies.^{14,15} Valacyclovir is comparable to famciclovir in terms of efficacy for reducing acute pain and accelerating healing.²⁰ As compared with acyclovir, valacyclovir and famciclovir require fewer daily doses, but are more expensive. While controlled trials have initiated treatment within 72 hours of onset of the rash, and treatment is recommended to start within this interval and as early as possible, many experts recommend that if new skin lesions are still appearing or complications of herpes zoster are present, treatment should be initiated even if the rash began more than 3 days prior. Treatment is usually given for 7 days in the absence of complications of herpes zoster. Intravenous acyclovir is recommended for immunocompromised persons who require hospitalization or for persons with severe neurologic complications. Foscarnet is used for immunocompromised patients with acyclovir-resistant VZV.

Corticosteroids—The use of corticosteroids with antiviral therapy for uncomplicated herpes zoster remains controversial. Randomized controlled trials have demonstrated

benefits of a tapering course of prednisone²¹ or prednisolone¹², including reducing acute pain,^{12,21} improving performance of activities of daily living,²¹ accelerated early healing,¹² and time to complete healing in one²¹ but not another study.¹² The addition of corticosteroids to antiviral therapy has not been shown to reduce the incidence of PHN. Owing to their immunosuppressive properties, corticosteroids should not be administered for herpes zoster without concomitant antiviral therapy. Corticosteroids should generally be avoided in patients with hypertension, diabetes mellitus, peptic ulcer disease, or osteoporosis; particular caution is warranted in elderly patients, who are at increased risk of serious adverse events with treatment. Prednisone is used for treatment of certain CNS complications of herpes zoster such as vasculopathy or Bell's palsy in non-immunocompromised patients.

Acute pain associated with herpes zoster—Several medications have been used for the treatment of acute pain associated with herpes zoster (Table 4). Nonsteroidal anti-inflammatory drugs or acetaminophen can be tried for patients with mild pain. Opioids, such as oxycodone, are used for more severe pain associated with herpes zoster. Opioids were more effective than gabapentin for herpes zoster pain in a randomized, placebo-controlled trial.²² Gabapentin was shown in one,²³ but not another²² controlled trial to reduce pain associated with herpes zoster. Lidocaine patches reduced pain associated with herpes zoster in a placebo-controlled trial, and should only be applied to intact skin, not to the area of the rash.²⁴ While tricyclic antidepressants have not been shown to be effective in randomized controlled clinical trials for acute zoster pain, they have been used when opioids are insufficient for pain.

Eye disease associated with zoster—Patients with zoster in the V1 distribution of the trigeminal nerve (including lesions on the forehead and the upper eyelid) and either the tip of the nose or new visual symptoms should be evaluated by an ophthalmologist. Other treatment may be needed in addition to antiviral therapy, including mydriatics to dilate the pupil and reduce the risk of scarring (synechiae), topical corticosteroids for keratitis, episcleritis, or iritis, medications to reduce ocular pressure for glaucoma, or intravitreal antiviral therapy for immunocompromised patients with retinal necrosis.

Postherpetic neuralgia—Pain associated with PHN is often challenging to treat. Detailed discussion of PHN management is beyond the scope of this article. In brief, medications demonstrated in randomized trials to reduce pain associated with PHN include topical lidocaine²⁵, anticonvulsants (e.g. gabapentin²⁶ or pregabalin²⁷), opioids,²⁸ tricyclic antidepressants (e.g. nortriptyline²⁹), and capsaicin³⁰. Combined modalities, such as gabapentin and nortriptyline³¹ or an opiate and gabapentin³² have been more effective for PHN than single agent therapy, but also confer greater risk of side effects. Even with treatment, many patients do not have adequate relief of pain and referral to a pain specialist can be helpful.

Prevention of Herpes Zoster—A live attenuated zoster vaccine is recommended by the Advisory Committee on Immunization Practices in persons age 60 and older to prevent herpes zoster and its complications, including PHN.^{9,33} Based on the results of a recent

clinical trial, the vaccine is now approved by the FDA for use in persons age 50 or above to prevent herpes zoster.³⁴ The efficacy of the vaccine to prevent herpes zoster is 70% in persons aged 50 to 59, 64% in persons aged 60 to 69, and 38% in persons 70 or above.^{33–35} Vaccine efficacy for prevention of PHN is 66% in persons aged 60 to 69 and 67% in persons 70 or older.^{33,35} A follow-up study showed that the reduction in the risk of herpes zoster remained significant for at least 5 years after vaccination, though vaccine effectiveness declined over time.³⁶ Among vaccinated (as compared with unvaccinated) persons who develop herpes zoster, pain was significantly shorter in duration and less severe.³³ The vaccine can be given to persons with a prior history of herpes zoster; in a recent report, rates of adverse events associated with vaccination were similar between those who had prior zoster (mean 3.6 years prior to vaccination) and those with no history of zoster.³⁷

The optimal timing of vaccination after an episode of zoster is uncertain. Since the risk of recurrent zoster after a recent episode of the disease is relatively low,³⁸ and the cellular immune response to VZV is similar during the first 3 years after receiving the zoster vaccine and after having an episode of zoster,³⁹ one might delay vaccination in immunocompetent persons with a recent history of zoster providing that it is well documented by a health care provider. The vaccine is contraindicated in persons with hematologic malignancies whose disease is not in remission or have received cytotoxic chemotherapy within 3 months, in persons with T cellular immunodeficiency (e.g. HIV infection with CD4 count $\leq 200/\text{mm}^3$ or $<15\%$ of total lymphocytes), and in those receiving high dose immunosuppressive therapy (e.g. ≥ 20 mg of prednisone daily for ≥ 2 weeks or anti-TNF therapy).

Infection Control—While herpes zoster is less contagious than varicella, patients with zoster can transmit VZV to susceptible persons and cause varicella. Non-immunocompromised persons with dermatomal zoster should be placed on contact precautions and lesions should be covered if possible;⁴⁰ however, virus transmission has occasionally been reported in such patients.⁴¹ Persons with disseminated lesions or immunocompromised persons with herpes zoster should be placed on airborne and contact precautions until all lesions have crusted.

AREAS OF UNCERTAINTY

Improved therapies are needed for pain associated with herpes zoster and PHN and to prevent development of PHN after herpes zoster. Studies are needed to determine which patients are at highest risk for PHN so that more aggressive therapy can be given. There is uncertainty regarding the effectiveness of the zoster vaccine in persons ≥ 80 years old, the safety and effectiveness of the vaccine in persons with immunocompromising conditions that are currently considered contraindications to vaccination, the duration of immunity induced by the vaccine, and whether booster doses will be needed.

GUIDELINES

Recommendations have been published for the management of herpes zoster from a group of experts⁸ and for prevention of herpes zoster by the Advisory Committee on Immunization Practices.⁹ The present review is generally concordant with these recommendations.

CONCLUSIONS AND RECOMMENDATIONS

Whereas herpes zoster is often mild in healthy young persons, older individuals are at increased risk of pain and complications including PHN, ocular disease, motor neuropathy, or CNS disease. In the vast majority of cases, the diagnosis can be made clinically. Antiviral therapy is most beneficial for persons with complications of zoster or at increased risks of complications such as older persons and immunocompromised persons, and should be initiated as soon as possible, and generally within 72 hours of the onset of rash. Valacyclovir or famciclovir are preferred over acyclovir due to reduced frequency of dosing and higher levels of antiviral drug activity. The patient described above should receive oral antiviral therapy, medication for pain (e.g. opioid, with addition of gabapentin if needed), and prompt referral to an ophthalmologist. He should also be advised to avoid contact with persons who have not had varicella or have not received the varicella vaccine until his lesions have completely crusted. I would recommend herpes zoster vaccination in the future to reduce his risk of recurrence, but would tend to defer this in an immunocompetent patient for approximately 3 years, given that a zoster episode is expected to boost his cellular immune response to VZV.

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Take Home Points

In the absence of zoster vaccine, persons who live to age 85 have a 50% chance of developing herpes zoster.

Persons most likely to benefit from antiviral therapy for herpes zoster include those with complications of zoster, and those at risk for complications including immunocompromised persons, those ≥ 50 years old, and those with severe pain or severe rash.

Antivirals hasten resolution of herpes zoster lesions and decrease severity of acute pain, but have not been convincingly been shown to reduce the risk of postherpetic neuralgia.

Valacyclovir or famciclovir are preferred to acyclovir due to ease of dosing and higher levels of antiviral drug activity.

Patients with zoster and new visual symptoms should be evaluated by an ophthalmologist to determine whether ocular-specific therapy is needed.

The zoster vaccine is approved by the FDA for persons ≥ 50 years old and is used in persons with and without a history of zoster.

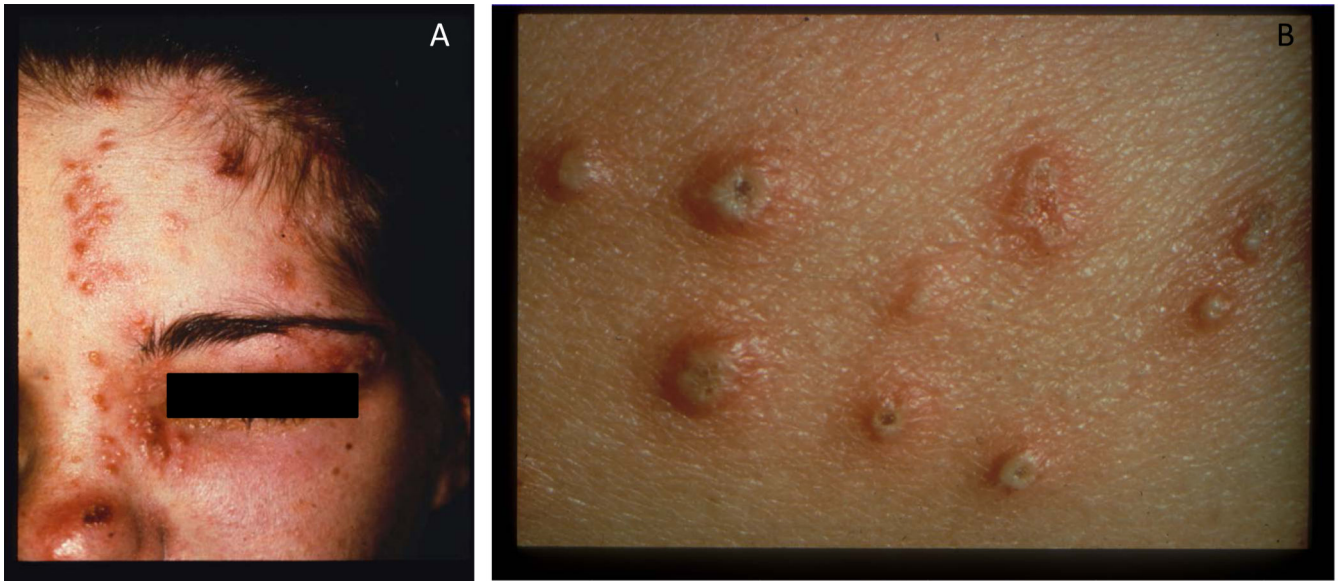


Figure 1.

(A) Zoster in the ophthalmic (V1) branch of the trigeminal ganglia. Photograph courtesy of Michael Oxman, M.D. (B) Vesicles and pustules in a patient with zoster. These are representative photographs and are not from the case presented.

Table 1

Selected Complications of Herpes Zoster in the Non-Immunocompromised Host

Complication	Manifestations	Site of VZV reactivation
Aseptic meningitis	Headache, meningismus	Cranial nerve V
Bacterial superinfection	Streptococcus, staphylococcus cellulitis	Any sensory ganglia
Bell's palsy	Unilateral facial paralysis	Cranial nerve VII
Eye involvement (herpes zoster ophthalmicus)	Keratitis, episcleritis, iritis, conjunctivitis, uveitis, acute retinal necrosis, optic neuritis, acute glaucoma.	Cranial nerve II, III, or V (ophthalmic [V1] branch)
Hearing impairment	Deafness	Cranial nerve VIII
Motor neuropathy	Weakness, diaphragm paralysis, neurogenic bladder	Any sensory ganglia
Postherpetic neuralgia	Pain persisting for 90 days after onset of rash	Any sensory ganglia
Ramsay-Hunt syndrome	Ear pain and vesicles in the canal, numbness of anterior tongue, facial paralysis	Cranial nerve VII geniculate ganglia and spread to cranial nerve VIII
Transverse myelitis	Paraparesis, sensory loss, sphincter impairment	Vertebral ganglia
Vasculopathy (encephalitis)	Vasculitis of cerebral arteries, confusion, seizures, TIAs, stroke	Cranial nerve V

Table 2Indications for Antiviral Treatment for Patients with Herpes Zoster^a

≥50 years old
Moderate or severe pain
Severe rash
Involvement of the face or eye
Other complications of zoster
Immunocompromised persons

^a While antivirals may benefit additional patients with zoster, they are primarily recommended by experts for these 6 groups who either have complications or who are at increased risk of complications from zoster.^{8,9}

Table 3

Antiviral Therapy for Herpes Zoster*

Medication	Dose	Effects Observed in Controlled Trials	Side Effects
Nonimmunocompromised persons			
Acyclovir (oral)	800 mg po 5 times daily for 7–10 days	Reduced time to last new lesion formation, loss of vesicles, full crusting, cessation of viral shedding; reduced severity of acute pain ^{10–12}	malaise
Famciclovir (oral)	500 mg po 3 times daily × 7 days	Reduced time to last new lesion formation, loss of vesicles, full crusting, cessation of viral shedding, cessation of pain ^{13,14}	headache, nausea
Valacyclovir (oral)	1 gm po 3 times daily × 7 days	Reduced time to last new lesion formation, loss of vesicles, full crusting, cessation of pain ^{15,16}	headache, nausea
Immunocompromised persons requiring hospitalization or severe neurologic complications			
Acyclovir (intravenous)	10 mg/kg iv every 8 hrs × 7–10 days	Reduced time to last new lesion formation, full crusting, cessation of virus shedding, cessation of pain; reduced cutaneous dissemination, reduced visceral zoster ^{17,18}	renal insufficiency
Foscarnet ⁺ (for acyclovir-resistant VZV)	40 mg/kg every 8 hr until healed	not reported	renal insufficiency, hypokalemia, hypocalcemia, hypomagnesemia, hypophosphatemia, nausea, diarrhea, vomiting, anemia, granulocytopenia, headache

* All drugs in this table are available as generics

⁺ Not approved for this use by FDA

Table 4

Medications commonly used for treatment of acute pain associated with herpes zoster*

Medication	Dose	Titration	Maximum Dose	Side Effects
Analgesics-opioid and non-opioid				
Oxycodone	5 mg every 4 hrs as needed	Increase by 5 mg 4 times daily every 2 days as tolerated	None specified, but do not exceed 120 mg daily without pain specialist	Drowsiness, dizziness, constipation, nausea, vomiting
Tramadol	50 mg once or twice daily	Increase by 50–100 mg daily in divided doses every 2 days as tolerated	400 mg daily; 300 mg daily if >75 years old	Drowsiness, dizziness, constipation, nausea, vomiting
Corticosteroids[^]				
Prednisone	60 mg daily for 7 days, then decrease to 30 mg daily for 7 days, then decrease to 15 mg daily for 7 days	None	60 mg	Gastrointestinal distress, nausea, vomiting, mood changes, edema, glucose intolerance, increased blood pressure
Anticonvulsants				
Gabapentin	300 mg at bedtime or 100–300 mg 3 times daily	Increase by 100–300 mg 3 times daily every 2 days as tolerated	3600 mg daily	Drowsiness, dizziness, ataxia, peripheral edema
Pregabalin	75 mg at bedtime or 75 mg twice daily	Increase by 75 mg twice daily every 3 days as tolerated	600 mg daily	Drowsiness, dizziness, ataxia, peripheral edema
Tricyclic antidepressants				
Nortriptyline	25 mg at bedtime	Increase by 25 mg daily every 2–3 days as tolerated	150 mg daily	Drowsiness, dry mouth, blurred vision, weight gain, urinary retention
Topical therapy				
Lidocaine 5% patch	One patch topically for up to 12 hours to intact skin only	None	One patch for up to 12 hours within a 24-hour period	Local irritation; if systemic absorption can cause drowsiness, dizziness

* Modified from Dworkin RH, Johnson RW, Breuer J, et al. Recommendations for the management of herpes zoster. Clin Infect Dis. 2007;44 Suppl 1:S1–26, by permission of Oxford University Press.

This list provides examples and is not meant to be comprehensive

[^] The use of corticosteroids is controversial as they are often poorly tolerated in older patients.